Gabriela Mendoza

+1 (786) 520-8317 – Gmendo1805@gmail.com – Mendog12@my.erau.edu – Daytona Beach, FL
linkedin.com/in/gabriela-mendoza

EXECUTIVE SUMMARY

Mechanical Engineering student concentrating in **Robotics and Autonomous Systems**. Three production operations co-ops at Gulfstream, across wing box assembly, test, and completions. I like being close to the build, designing a part, testing it, and seeing it come together on the floor. Looking for a role in **R&D, production, or test engineering** where I can stay hands on and work on continuous improvement.

EDUCATION

Bachelor of Science in Mechanical Engineering — May 2028
Embry-Riddle Aeronautical University — Daytona Beach, FL
Area of Concentration: Robotics and Autonomous Systems | Minor: CAD/CAM — GPA 3.1

PROJECT EXPERIENCE

DARPA Heavy-Lift Drone Project — May 2026 – August 2026
Maxweights (Drone Start-Up)

- Developed a heavy-lift drone designed to complete a **5-nautical-mile mission carrying a minimum 110 lb payload** within a 55 lb aircraft weight limit
- Supported hands-on assembly and integration of airframe, propulsion, and avionics components under weight constraints
- Gained hands-on experience with large-scale drone flight controls, sensor calibration, and system integration
- Performed mechanical and electrical troubleshooting to improve flight performance and reliability

Women's Baja SAE – Off-Road Vehicle Team — Fall 2023 – Fall 2024
Embry-Riddle Aeronautical University — Daytona Beach, FL

- Gained hands-on experience with the assembly of off-road vehicles for intercollegiate competition
- Practiced TIG welding techniques to create strong, clean welds for vehicle components
- Helped new members improve their welding skills by sharing tips and techniques while working closely with experienced team members

INTERNSHIP EXPERIENCE

Production Operations Engineering Co-op – Completions — May 2026 – July 2026
Gulfstream Aerospace Corporation — Savannah, GA

- Investigated engineering, manufacturability, and ergonomic installation issues on the floor, including electrical routing and component interference, and brought findings to design engineering for continuous improvement
- Assigned to a build team and a specific aircraft, following it through the production phase and reviewing engineering changes, open discrepancies, work orders, and part shortages to catch issues before they affected the build
- Traced **55–80 labor hours** per build to wiring and component failures in cabin furniture, then started a test box project to catch them before install

Production Operations Engineering Co-op – Test Station — August 2025 – December 2025
Gulfstream Aerospace Corporation — Savannah, GA

- Investigated unintended 406 MHz ELT antenna activations and evaluated mitigations including electromagnetic shielding, RF-to-voltage LED notification, and an in-house monitoring solution
- Designed and built an adjustable test rig replicating aircraft brake valve hydraulic tube configurations to evaluate vibration-induced noise and validate alternative support configurations
- Completed a floor rotation with test specialists and mechanics, executing test specifications to validate aircraft systems

Production Operations Engineering Co-op – Wing Box Assembly — January 2025 – May 2025
Gulfstream Aerospace Corporation — Savannah, GA

- Analyzed part request data in Excel to identify 10+ components repeatedly pulled without planned demand, surfacing hidden manufacturing challenges
- Created and implemented 10 CATIA-based shop aids that improved work instruction clarity, reduced build time, and prevented recurring assembly errors

SKILLS

Engineering Software: CATIA V5, SOLIDWORKS, Fusion 360, MATLAB, SAP, Solumina, SmarTeam, ANSYS APDL
Manufacturing: TIG welding, soldering, electrical integration, shop aid development, Haas CNC mill (beginner)
Certification: Catia Design Fundamentals with SmarTeam, MATLAB Onramp
Languages: Native fluency in English and Spanish

LEADERSHIP & ACTIVITIES

- Member, Women's Baja SAE and Society of Hispanic Professional Engineers (SHPE) – Embry-Riddle Aeronautical University
- President, Tour Ambassador Program; Captain, Women's Varsity Soccer Team – Divine Savior Academy